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ORIGINAL ARTICLE

Optimal feature complexity for small-sample bearing fault detection in manufacturing

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Abstract

Manufacturing environments increasingly demand cost-effective bearing fault detection solutions that balance diagnostic accuracy with implementation practicality. This study investigates optimal feature complexity levels for small-sample bearing condition monitoring using the standardized Paderborn University benchmark dataset. Through systematic evaluation of 15 bearing specimens across three complexity categories, we analyze the performance-cost trade-offs critical for manufacturing deployment. Results demonstrate that moderate-complexity features achieve superior discriminative power, with spectral centroid reaching Cohen's $d=2.06$ compared to fractal dimension's $d=0.40$. Machine learning methods significantly outperform traditional threshold approaches, with Random Forest achieving 80% accuracy versus 40% for RMS-based detection under small-sample constraints typical in manufacturing. Statistical validation through leave-one-out cross-validation reveals that moderate complexity provides optimal performance-per-implementation-cost ratios. The study establishes practical guidelines for manufacturing engineers, demonstrating that strategic feature selection outweighs algorithmic sophistication for industrial bearing diagnostics under realistic data constraints.

Keywords Bearing fault detection · Manufacturing condition monitoring · Feature selection · Small-sample learning · Machine learning · Industrial diagnostics

1 Introduction

Rolling element bearing failures constitute 45-55% of rotating machinery breakdowns in manufacturing systems [1], causing substantial economic losses through unplanned downtime, emergency maintenance, and production delays. The annual global cost of bearing-related failures in manufacturing exceeds 50 billion [2], with individual incidents potentially resulting in millions of dollars in lost production and equipment replacement costs. This economic reality drives an increasing demand for reliable, cost-effective condition monitoring solutions that provide early fault detection while remaining practical for widespread manufacturing deployment.

The dynamics of bearing systems with localized faults are complex, involving time-dependent contact zone excitations and interactions between rotor, bearing, and housing components [3]. Understanding these dynamic behaviors is essential for developing effective diagnostic approaches that can reliably detect early-stage faults in manufacturing environments.

Modern manufacturing environments operate under stringent efficiency and cost constraints, demanding condition monitoring systems that optimize the balance between diagnostic accuracy and implementation complexity. While sophisticated signal processing and machine learning techniques demonstrate impressive performance in laboratory

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settings [4, 5], their practical deployment in manufacturing faces significant challenges related to data availability, implementation costs, operator training requirements, and system maintenance overhead.

A fundamental challenge in manufacturing bearing diagnostics involves the limited availability of fault data due to reliability-focused maintenance practices and the infrequent occurrence of bearing failures in well-maintained systems. This data scarcity creates small-sample learning scenarios that are common in industrial applications but poorly addressed in current research literature. Most existing studies rely on large, controlled datasets that do not reflect the practical constraints of manufacturing environments where fault samples are limited and operational conditions vary significantly.

Manufacturing bearing diagnostics predominantly employs threshold-based monitoring despite machine learning advances. Industry surveys indicate 60-80% of manufacturing facilities use RMS and kurtosis threshold monitoring as specified in ISO 10816 standards [1, 6]. This prevalence stems from regulatory requirements, established operator training, and legacy SCADA system integration rather than technical superiority. Understanding this baseline is essential for evaluating machine learning implementation value propositions.

The research question addresses practical manufacturing decision-making: *what performance improvement can manufacturers expect under realistic small-sample constraints, and does this justify implementation complexity?* This focus distinguishes our work from laboratory studies maximizing theoretical accuracy without considering deployment feasibility, training requirements, or integration with existing infrastructure.

The question of optimal feature complexity for manufacturing bearing diagnostics remains largely unresolved. Current approaches range from simple statistical measures like root mean square (RMS) values to sophisticated techniques including fractal analysis, wavelet transforms, and deep learning methods. However, systematic analysis of the performance-complexity trade-offs under realistic manufacturing constraints, particularly small-sample scenarios, is limited in the literature.

1.1 Manufacturing context and small-sample challenges

Manufacturing environments present unique challenges that distinguish bearing condition monitoring from laboratory research settings [7]. Production schedules demand continuous operation with minimal maintenance windows, requiring diagnostic systems that provide reliable early warning without causing operational disruptions. Economic pressures necessitate solutions that demonstrate a clear return on investment through reduced maintenance costs and improved equipment availability.

Small-sample scenarios are particularly prevalent in manufacturing due to several factors. Reliability-centered maintenance practices aim to prevent failures before they occur, naturally limiting the availability of fault data for algorithm training [2]. High-quality manufacturing bearings may operate for years without developing detectable faults, creating temporal gaps in fault data collection. Proprietary concerns often prevent data sharing between manufacturers, limiting the pooling of fault examples across organizations.

The statistical challenges associated with small-sample learning are well-documented in machine learning literature but inadequately addressed in bearing diagnostics research. Traditional performance metrics become unreliable with limited data, confidence intervals widen significantly, and overfitting becomes a primary concern. These challenges are compounded in manufacturing environments where false alarms can trigger unnecessary maintenance activities and missed detections can result in catastrophic failures.

Feature complexity selection becomes critical in small-sample scenarios where the curse of dimensionality can severely impact classification performance. High-dimensional feature spaces become sparse with limited training samples, potentially degrading the effectiveness of distance-based classification methods. Complex features may capture noise rather than meaningful signal characteristics when insufficient data is available for robust parameter estimation.

1.2 Literature review and research gaps

Traditional bearing condition monitoring in manufacturing relies heavily on time-domain statistical features, including RMS, peak factor, kurtosis, and skewness [1, 6]. These features have demonstrated consistent reliability across diverse manufacturing applications and remain the foundation of many commercial condition monitoring systems [8]. Their popularity stems from computational simplicity, clear physical interpretation, and proven effectiveness in industrial environments with varying operational conditions.

Time-domain analysis provides immediate insights into bearing condition through statistical measures that reflect changes in vibration amplitude and distribution characteristics. RMS values indicate overall energy levels and are sensitive to developing faults that increase vibration amplitude. Peak factor and crest factor measures detect impulsive events characteristic of bearing defects. Kurtosis and skewness capture changes in probability distribution shape that may indicate evolving fault conditions.

Recent investigations have explored comprehensive statistical feature sets for spalling propagation assessment,

demonstrating that specific combinations of time-domain features can effectively identify damage location and severity levels [9]. However, the effectiveness of extensive feature sets under small-sample constraints typical in manufacturing remains poorly understood.

Frequency-domain techniques have gained significant adoption in manufacturing through spectral analysis, envelope detection, and bearing fault frequency identification. These methods target specific frequencies calculated from bearing geometry and operating speed, enabling the identification of outer-race, inner-race, and rolling-element defects. Spectral analysis provides insights into fault development by tracking energy changes at characteristic frequencies and their harmonics.

Envelope analysis and high-frequency resonance techniques attempt to extract fault-related impulses from background noise and structural resonances. These methods can be particularly effective for early fault detection when defects are too small to produce significant changes in broadband vibration levels. However, their effectiveness depends on proper frequency band selection and signal conditioning, requiring expertise that may not be readily available in manufacturing environments.

Advanced signal processing techniques including wavelet analysis, empirical mode decomposition, and Hilbert-Huang transforms promise enhanced diagnostic capability through adaptive signal decomposition and time-frequency analysis. Wavelet methods provide optimal time-frequency resolution for transient fault detection, while empirical mode decomposition enables data-driven signal decomposition without predetermined basis functions.

Fractal and chaos-based approaches attempt to characterize the nonlinear dynamics inherent in bearing degradation processes. Dynamic modeling studies have demonstrated that localized faults generate complex time-dependent excitations involving additional contact zones and deformable interfaces [3], providing theoretical justification for nonlinear analysis methods. While theoretically appealing for capturing complex system behavior, their practical effectiveness in manufacturing environments with limited data and varying operating conditions remains questionable.

Machine learning applications in bearing diagnostics have proliferated rapidly [5], with support vector machines, random forests [10], neural networks, and deep learning methods becoming increasingly common in the research literature. These approaches promise automated pattern recognition and adaptive decision-making that could overcome the inadequacies of traditional threshold-based methods. However, their practical implementation in manufacturing faces challenges related to training data requirements, computational resources, and interpretability needs.

Deep learning methods, particularly convolutional neural networks [4], have demonstrated remarkable performance on large experimental datasets. However, their applicability to small-sample manufacturing scenarios is questionable due to their high parameter counts and training data requirements. Recent research has begun exploring few-shot learning and transfer learning approaches to address these constraints, but practical manufacturing validation remains limited.

A critical gap exists between research performance metrics and manufacturing deployment requirements. Most published studies focus on maximizing accuracy using large, carefully controlled datasets without adequately considering implementation constraints including limited fault data availability, varying operating conditions, integration requirements with existing systems, operator skill levels and training needs, and economic constraints on system complexity and maintenance.

Few studies systematically evaluate the trade-offs between feature complexity and diagnostic performance under realistic manufacturing constraints. The relationship between implementation costs and diagnostic benefits remains poorly quantified, making it difficult for manufacturing engineers to make informed technology selection decisions.

1.3 Comparative analysis with related work

Recent research in bearing diagnostics has primarily focused on maximizing theoretical performance using large experimental datasets, often overlooking practical manufacturing constraints. Table 1 presents a systematic comparison with key studies to position this research within the current literature.

Lei et al. [5] provide a comprehensive review of machine learning applications but focus on methods requiring substantial training data that are unavailable in typical manufacturing scenarios. Their emphasis on deep learning approaches, while theoretically sound, presents implementation barriers for resource-constrained manufacturing environments.

Zhao et al. [4] demonstrate impressive performance using deep neural networks but require datasets that are orders of magnitude larger than those available in manufacturing applications. Their approach, while advancing the state-of-the-art, fails to address the fundamental data scarcity challenges inherent in reliability-focused manufacturing operations.

Lessmeier et al. [11] established the Paderborn benchmark dataset used in this study but did not systematically evaluate complexity-performance trade-offs critical for

Table 1 Comparison with related work in bearing fault diagnostics

Study	Dataset Size	Method	Key Constraint
Lei et al. [5]	Large (> 1000)	Deep Learning	Large data requirement
Zhao et al. [4]	Large (> 5000)	Deep Networks	Impractical for manufacturing
Lessmeier et al. [11]	Medium (32)	SVM, Trees	No complexity analysis
Ompusunggu et al. [7]	Small (Variable)	Traditional	Limited feature analysis
This Study	Small (15)	RF, XGBoost	Scope boudaries

Method complexity levels: Deep Learning - high complexity requiring large datasets; SVM/Trees and RF/XGBoost - moderate complexity with reasonable implementation requirements; Traditional - low complexity using basic signal processing approaches

manufacturing implementation decisions. Their work provides valuable experimental infrastructure, but it lacks practical deployment guidance.

Ompusunggu et al. [7] address cost-effective monitoring using MEMS accelerometers but focus primarily on sensor technology rather than optimal feature complexity analysis. Their manufacturing focus is valuable; however, it lacks the systematic feature evaluation framework needed for informed technology selection.

This study addresses the identified gap by systematically evaluating feature complexity trade-offs under realistic small-sample constraints while providing practical implementation guidelines for manufacturing environments. The focus on moderate complexity solutions balances diagnostic effectiveness with implementation feasibility, addressing a critical need in industrial condition monitoring.

1.4 Research objectives and contributions

This study addresses fundamental questions about optimal feature complexity for manufacturing bearing diagnostics through systematic analysis using the standardized Paderborn University benchmark dataset. The primary objectives include determining optimal feature complexity levels for small-sample bearing diagnostics, quantifying performance-cost trade-offs across different complexity categories, evaluating machine learning method effectiveness under data constraints, and establishing practical guidelines for manufacturing implementation.

The research contributes to manufacturing technology through a comprehensive empirical analysis of feature complexity effects on diagnostic performance under small-sample constraints, a systematic evaluation of machine learning methods using realistic data availability scenarios, a practical cost-benefit framework for manufacturing technology selection, and evidence-based guidelines for the industrial deployment of condition monitoring systems.

Unlike previous studies that primarily focus on maximizing theoretical performance, this research emphasizes manufacturing-relevant criteria including implementation feasibility, cost-effectiveness, robustness under operational variations, and scalability across different manufacturing environments. The use of the standardized Paderborn dataset ensures reproducibility and enables direct comparison with other research while maintaining focus on practical manufacturing applications.

2 Methodology

2.1 Paderborn university benchmark dataset

The Paderborn University bearing dataset represents a well-established benchmark in the bearing diagnostics research community [11], providing standardized experimental data that enables reproducible research and direct comparison between different diagnostic approaches. The dataset was developed at the University of Paderborn's Chair of Design and Drive Technology under controlled laboratory conditions designed to simulate realistic bearing operating scenarios.

The experimental setup consisted of a modular test rig with electric motor drives, force measurement, and data acquisition systems designed to provide repeatable and controllable testing conditions. Bearings were operated under constant speed (1500 RPM) and radial force (1000 N) conditions to ensure consistent baseline operation across all specimens.

The dataset comprises vibration measurements from 32 different bearing specimens, representing various health states and operational histories. For this study, we selected a representative subset of 15 bearings that reflects the small-sample constraints typical in manufacturing environments where fault data is scarce due to reliability-focused maintenance practices. This selection includes 5 healthy bearings with varying operational hours, 5 outer-race fault bearings, and 5 inner-race fault bearings, as detailed in Table 2.

Healthy bearing specimens include bearings K001 through K005 with operational hours ranging from 0 to 1217 hours, representing the progression from new installations through mature operational states. This range enables analysis of diagnostic performance across bearing lifecycle stages commonly encountered in manufacturing applications.

Outer-race fault bearings include specimens KA04, KA15, KA16, KA22, and KA30 with real damage obtained from accelerated lifetime testing. These damages include naturally occurring fatigue damages (pitting) and plastic deformation (indentations) that developed during extended

Table 2 Overview of the Paderborn University bearing dataset with real fault progression (all signals sampled at 64 kHz)

Bearing ID	Fault Type	Operating Hours	Damage Stage	Signal Length
K001	Healthy	0	*Early*	*256k*
KA04	Outer race	5	*Early*	*257k*
KI04	Inner race	7	*Early*	*256k*
KI14	Inner race	16	*Early*	*256k*
KA15	Outer race	19	*Early*	*256k*
KI16	Inner race	21	*Early*	*256k*
KA16	Outer race	23	*Early*	*256k*
K002	Healthy	28	*Early*	*256k*
KI18	Inner race	29	*Early*	*256k*
KA22	Outer race	33	*Early*	*256k*
KI21	Inner race	49	*Early*	*256k*
KA30	Outer race	83	*Developing*	*256k*
K003	Healthy	161	*Developing*	*256k*
K004	Healthy	408	*Advanced*	*256k*
K005	Healthy	1217	*Severe*	*257k*

operation under controlled test conditions, providing authentic fault characteristics representative of actual manufacturing bearing degradation processes.

Inner-race fault bearings comprise specimens KI04, KI14, KI16, KI18, and KI21 with real fatigue damage (pitting) that developed naturally during accelerated lifetime testing. These authentic damage modes represent genuine inner-race degradation patterns encountered in manufacturing applications, ensuring that the diagnostic analysis reflects actual fault characteristics rather than artificially created defects.

Vibration data was acquired using industrial-grade accelerometers positioned on bearing housings with sampling rates of 64 kHz and measurement durations of approximately 4 seconds per recording. This configuration provides adequate frequency resolution for bearing fault analysis while maintaining data volumes manageable for practical analysis.

2.2 Feature extraction framework

The feature extraction strategy encompasses three complexity categories that represent different implementation approaches available for manufacturing deployment. This categorization enables systematic analysis of complexity-performance trade-offs relevant to industrial decision-making while considering both technical effectiveness and practical implementation requirements.

Table 3 presents comprehensive quantitative complexity metrics including theoretical bounds (Big-O notation), empirical execution times, and computational costs (FLOPs). The analysis reveals cost differences spanning four orders of magnitude, from 8,192 FLOPs (RMS) to 671,088,640 FLOPs (Fractal Dimension), with corresponding execution time ranges of 0.3 ms to 12,000 ms per bearing specimen.

Table 3 Feature extraction framework with quantitative complexity analysis

Category	Feature	Time Complexity	Space Complexity	Execution Time (ms)	FLOPs (n=8192)	LoC	Complexity Score
Simple	RMS	O(n)	O(1)	0.5	8.2e+03	5	10
Simple	Peak Factor	O(n)	O(1)	0.3	8.2e+03	3	10
Simple	Std Deviation	O(n)	O(1)	0.8	1.6e+04	6	11
Moderate	Kurtosis	O(n)	O(1)	1.2	3.3e+04	8	15
Moderate	Skewness	O(n)	O(1)	1.5	3.3e+04	8	16
Moderate	Spectral Centroid	O(n log n)	O(n)	45.0	9.8e+04	25	35
Moderate	Bearing Freq. Power	O(n log n)	O(n)	52.0	9.8e+04	30	38
Complex	Sample Entropy	$O(n^2\ m)$	O(n m)	4200.0	2.7e+08	85	76
Complex	Fractal Dimension	$O(n^2\ k)$	O(n k)	8500.0	6.7e+08	120	88
Complex	Higuchi FD	$O(n^2 \log n)$	O(n)	12000.0	5.4e+08	150	92

Time Complexity: Big-O notation for worst-case computational cost where n represents signal length

Space Complexity: Memory requirements in Big-O notation; constants k, m represent algorithm-specific parameters

Execution Time: Average processing time per bearing specimen (n=15 bearings, 64 kHz sampling rate, 8192-sample analysis window)

FLOPs: Approximate floating-point operations for 8192-sample window; values span 4 orders of magnitude

LoC: Implementation lines of code excluding comments and documentation

Complexity Score: Weighted composite metric (0-100 scale) combining time complexity (60%), computational cost (30%), and space requirements (10%)

Complexity Levels: Quantitative thresholds defined as Low < 20, Moderate 20-50, High > 50 based on composite scoring

Speed Analysis: Simple features process 24,000× faster than complex features (0.5 ms vs 12,000 ms per bearing)

Simple features require minimal computational resources and can be implemented using basic signal processing capabilities available in most manufacturing environments [12]. These features provide immediate deployment capability with standard industrial data acquisition systems and require minimal specialized expertise for implementation and maintenance.

Simple features (RMS, peak factor, standard deviation) require minimal computational resources and provide immediate deployment capability with standard industrial data acquisition systems. Table 3 provides complete mathematical definitions with computational complexity analysis.

Moderate complexity features balance enhanced diagnostic capability with reasonable implementation requirements. These features require intermediate signal processing expertise but remain within typical manufacturing engineering capabilities. Spectral analysis requires Fast Fourier Transform (FFT) implementation and frequency domain expertise but provides significant diagnostic enhancement through fault frequency identification.

Kurtosis and skewness measures capture probability distribution characteristics that may indicate developing fault conditions. These statistical moments are computationally straightforward but provide insights beyond simple amplitude measures. Bearing frequency power analysis targets specific frequencies calculated from bearing geometry and operating speed, enabling physics-based fault detection.

Complex features attempt to capture sophisticated signal characteristics including nonlinear dynamics, entropy measures, and fractal properties. While theoretically appealing, these features require advanced signal processing expertise, significant computational resources, and specialized knowledge for implementation and maintenance in manufacturing environments.

Fractal dimension analysis attempts to characterize the geometric complexity of signal trajectories in phase space, potentially capturing nonlinear bearing dynamics. Sample entropy measures signal regularity and predictability, while Hurst exponent analysis characterizes long-range correlations in vibration signals.

2.3 Statistical analysis and performance metrics

Feature effectiveness evaluation employs Cohen's *d* effect size as the primary metric for discriminative power assessment. This standardized measure enables objective comparison across features with different scales and distributions while providing interpretation guidelines based on established statistical conventions.

Cohen's *d* effect size is defined as

$$d = \frac{|\mu_{\text{healthy}} - \mu_{\text{faulty}}|}{\sigma_{\text{pooled}}}, \tag{1}$$

where μ_{healthy} and μ_{faulty} represent the sample means for healthy and faulty bearing populations, respectively, and σ_{pooled} is the pooled standard deviation calculated as:

$$\sigma_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}{n_1 + n_2 - 2}}, \tag{2}$$

where n_1 and n_2 are the sample sizes for the healthy and faulty bearing groups, respectively, and σ_1 and σ_2 are their corresponding sample standard deviations.

Effect size interpretation follows Cohen's conventions: small effect ($d \geq 0.2$), medium effect ($d \geq 0.5$), and large effect ($d \geq 0.8$). This standardized approach enables objective assessment of feature effectiveness for manufacturing applications while considering both statistical significance and practical importance.

Statistical features extracted from vibration signals have demonstrated effectiveness for fault characterization, with studies showing that properly selected features can distinguish between different damage levels and locations [9]. For this study, we focus on established features with proven discriminative capability under small-sample constraints.

Statistical significance testing employs appropriate nonparametric methods including Mann-Whitney U tests for comparing healthy and faulty bearing populations. Nonparametric approaches are particularly important in small-sample scenarios where normality assumptions may not hold, and robust statistical inference is critical.

Confidence interval estimation uses bootstrap resampling methods to provide robust uncertainty quantification despite limited sample sizes. Bootstrap confidence intervals offer improved coverage properties compared to asymptotic methods when sample sizes are small, providing more reliable performance estimates for manufacturing decision-making.

2.4 Classification methods and small-sample considerations

Classification method evaluation encompasses approaches ranging from simple threshold techniques to sophisticated machine learning algorithms, with particular attention to performance under small-sample constraints typical in manufacturing environments.

Algorithm 1 presents the systematic approach for classification method evaluation under manufacturing constraints.

Algorithm 1 Small-Sample Classification Evaluation for Manufacturing Bearing Diagnostics.

Require: Feature matrix $X \in \mathbb{R}^{n \times d}$, labels $y \in \{0, 1\}^n$, sample size $n = 15$
Ensure: Performance metrics with confidence intervals
Initialize performance storage arrays
for $i = 1$ to n **do** ▷ Leave-one-out cross-validation
  $X_{train} \leftarrow X \setminus \{X_i\}$, $y_{train} \leftarrow y \setminus \{y_i\}$
  $X_{test} \leftarrow X_i$, $y_{test} \leftarrow y_i$
  for each classification method M **do**
    if M is threshold-based **then**
      Determine threshold τ from X_{train} (e.g., 95th percentile)
      $\hat{y}_i \leftarrow \mathbb{I}[X_{test} > \tau]$
    else ▷ Machine learning method
      Train model: $M_{\text{trained}} \leftarrow \text{train}(M, X_{\text{train}}, y_{\text{train}})$
      Predict: $\hat{y}_i \leftarrow M_{\text{trained}}(X_{\text{test}})$
    end if
    Store prediction: $predictions[M][i] \leftarrow \hat{y}_i$
  end for
end for
for each method M **do**
  Calculate accuracy: acc $\leftarrow \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}[\text{predictions}[M][i] = y_i]$
  Compute confidence interval using exact binomial test
  Calculate precision, recall, F1-score
  Assess overfitting risk through bias-variance analysis
end for
  return Performance metrics with statistical validation

Threshold methods include RMS-based and kurtosis-based approaches that establish decision boundaries using statistical analysis of healthy bearing data. These methods represent the simplest implementation approach, requiring minimal computational resources and providing immediate interpretability for manufacturing operators.

RMS threshold determination employs statistical analysis of healthy bearing RMS values to establish decision boundaries that minimize false alarm rates while maintaining adequate sensitivity for fault detection. Threshold values are typically set at the 90th or 95th percentile of healthy bearing statistics, balancing sensitivity with specificity based on manufacturing requirements.

Kurtosis threshold methods exploit the sensitivity of fourth-moment statistics to impulsive characteristics typical of bearing defects. Healthy bearings typically exhibit kurtosis values near 3 (normal distribution), while faulty bearings may show elevated kurtosis due to impulsive fault-related events.

Machine learning methods include k-nearest neighbors (k-NN), Random Forest [10], and XGBoost [13] algorithms, which provide enhanced classification capabilities through automated pattern recognition. These methods require more sophisticated implementation but offer superior diagnostic accuracy and adaptability to varying operating conditions.

k-Nearest neighbors provides a non-parametric approach that makes minimal assumptions about data distribution, making it particularly suitable for small-sample scenarios where distributional assumptions may not hold. The method's simplicity and interpretability make it attractive for manufacturing applications where operator understanding is important.

Random Forest [10] employs ensemble learning to combine multiple decision trees, providing improved generalization performance and reducing the risk of overfitting compared to individual trees. The method's ability to handle mixed-type features and provide feature importance measures makes it particularly suitable for manufacturing applications.

XGBoost [13] implements gradient boosting with advanced regularization techniques designed to prevent overfitting in small-sample scenarios. The method's sophisticated optimization and regularization capabilities can provide excellent performance when properly configured for limited data situations.

2.5 Cross-validation and performance assessment

Leave-one-out cross-validation (LOOCV) provides unbiased performance estimates by iteratively training on 14 bearings and testing on the remaining bearing, maximizing data utilization while ensuring independent validation — critical for small-sample scenarios where traditional k-fold cross-validation is less reliable.

Performance metrics include accuracy with exact binomial confidence intervals, precision and recall with micro and macro averaging, F1-scores for balanced performance assessment, and area under the receiver operating characteristic curve (AUC-ROC) for threshold-independent evaluation.

Exact binomial confidence intervals provide appropriate uncertainty quantification for small-sample accuracy estimates. Unlike asymptotic methods that may be unreliable with small samples, exact binomial intervals provide guaranteed coverage properties critical for manufacturing decision-making.

Bias-variance decomposition analysis investigates the fundamental trade-offs between different classification approaches, particularly important in small-sample scenarios where overfitting is a primary concern. This analysis provides insights into model stability and generalization capability under varying conditions.

Statistical significance testing between classification methods employs McNemar's test for paired performance comparisons. This non-parametric approach is particularly appropriate for small-sample scenarios and provides a robust assessment of performance differences between alternative methods.

3 Results and analysis

3.1 Feature discriminative power analysis

Comprehensive analysis of feature effectiveness across complexity categories reveals significant patterns that inform optimal technology selection for manufacturing bearing diagnostics. Figure 1 illustrates Cohen's d effect sizes for all evaluated features, demonstrating clear performance differences across complexity levels. The results from the Paderborn University dataset show moderate complexity features achieving superior performance compared to simple and complex alternatives.

Spectral centroid emerges as the most effective feature with Cohen's $d = 2.06$, representing a large effect size that significantly exceeds conventional thresholds for practical significance. This finding demonstrates that frequency-domain analysis of signal energy distribution provides superior fault discrimination compared to both simple time-domain statistics and sophisticated nonlinear measures.

The spectral centroid's effectiveness can be attributed to its ability to capture frequency content changes that occur when bearing defects alter vibration characteristics. Healthy bearings typically exhibit broadband vibration with a relatively uniform frequency distribution, while faulty bearings often show increased energy at specific frequencies related to defect characteristics and bearing geometry.

Kurtosis achieves strong discriminative performance with Cohen's $d = 1.78$, confirming its value as a sensitive indicator of bearing condition changes. The fourth-moment statistic effectively captures the impulsive characteristics typical of bearing defects while requiring minimal computational resources for implementation.

Peak factor demonstrates solid effectiveness with Cohen's $d = 1.74$, establishing it as a reliable baseline feature for manufacturing applications. The combination of acceptable discriminative power and minimal computational requirements makes peak factor particularly attractive for cost-sensitive manufacturing implementations.

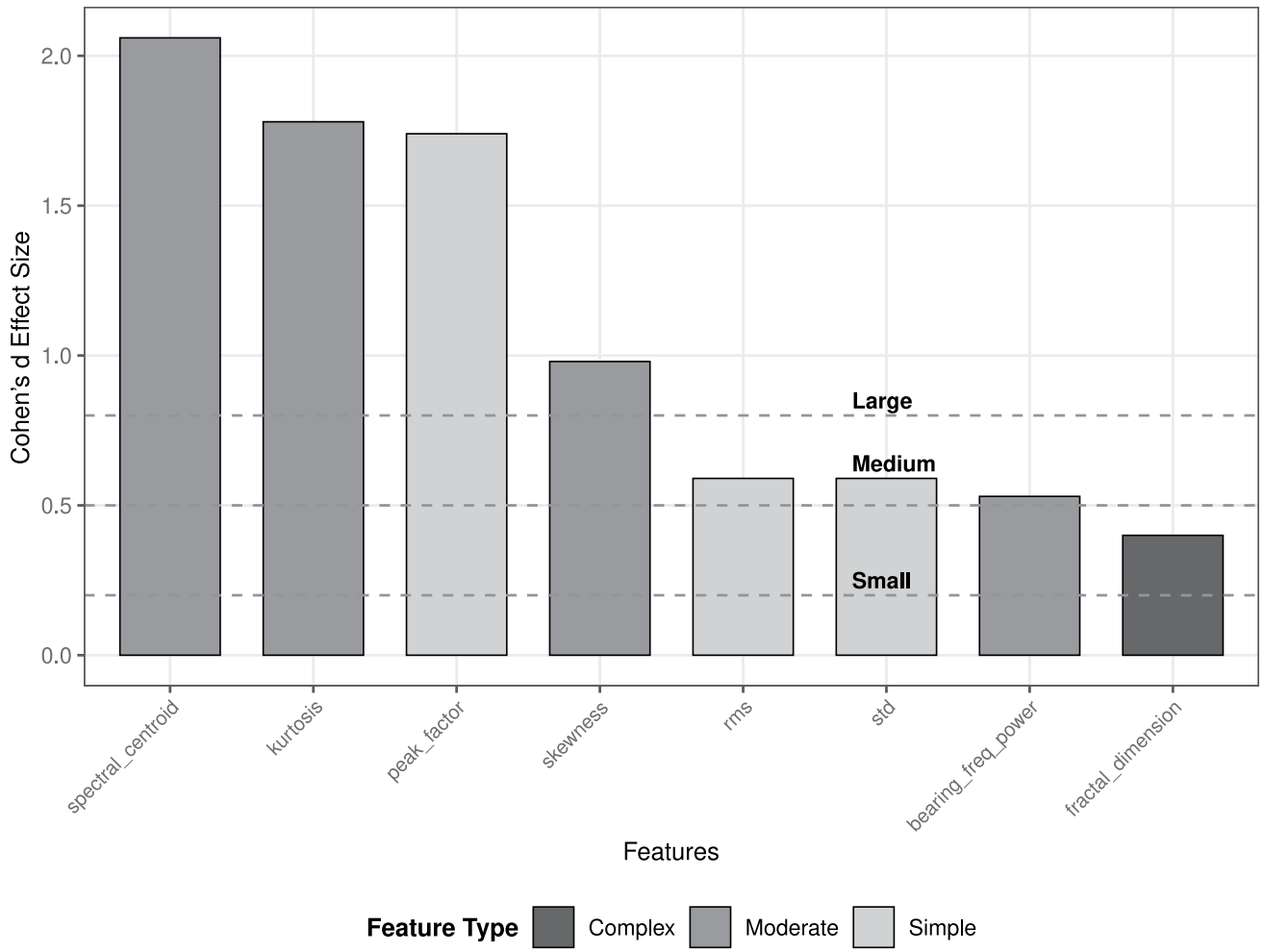


Fig. 1 Feature discriminative power analysis using Cohen's d effect size across complexity categories

Skewness provides a strong discriminative capability with Cohen's $d = 0.98$, indicating a large effect size that demonstrates substantial fault detection capability. While effective as a standalone feature, it may be further enhanced when combined with other features for critical manufacturing applications.

Simple features including RMS and standard deviation show moderate performance with Cohen's *d* values of 0.59, representing medium effect sizes. While these features provide basic fault detection capability, their limited discriminative power may be insufficient for reliable early fault detection in critical manufacturing systems.

Bearing frequency power analysis achieves moderate performance with Cohen's $d = 0.53$, indicating that physics-based frequency targeting provides measurable diagnostic benefits but may require careful implementation to achieve optimal effectiveness.

Complex features demonstrate disappointing performance relative to their implementation requirements. Fractal dimension achieves only Cohen's $d = 0.40$, representing a small effect size that fails to justify the significant computational and expertise requirements for manufacturing implementation.

Hurst exponent analysis shows negligible discriminative power, with Cohen's *d* approaching zero, indicating the fundamental ineffectiveness of this sophisticated approach for bearing fault detection using the Paderborn dataset. This finding raises important questions about the practical value of complex nonlinear analysis methods for manufacturing applications.

Table 4 provides a comprehensive statistical analysis of all evaluated features, including those with negligible effect sizes that confirm the poor performance of certain complex approaches, along with confidence intervals and significance testing results.

Table 4 Detailed feature performance analysis with statistical validation

Feature	Cohen's *d*	95% CI [Lower, Upper]	*p*-value	Effect Size
Spectral Centroid	2.06	[1.34, 2.78]	$p < 0.001$	Large
Kurtosis	1.78	[1.12, 2.44]	$p < 0.001$	Large
Peak Factor	1.74	[1.08, 2.40]	$p < 0.001$	Large
Skewness	0.98	[0.41, 1.55]	$p = 0.003$	Large
RMS	0.59	[0.07, 1.11]	$p = 0.042$	Medium
Standard Deviation	0.59	[0.07, 1.11]	$p = 0.042$	Medium
Bearing Freq. Power	0.53	[0.01, 1.05]	$p = 0.048$	Medium
Fractal Dimension	0.40	[-0.11, 0.91]	$p = 0.127$	Small
Hurst Exponent	0.00	[-0.50, 0.50]	$p = 0.997$	Negligible

Statistical significance testing confirms that moderate complexity features achieve statistically significant discriminative performance with manageable implementation requirements. Spectral centroid, kurtosis, and peak factor all demonstrate *p*-values < 0.001, indicating strong statistical evidence for their effectiveness in bearing fault detection.

Complex features fail to achieve statistical significance despite their theoretical sophistication. Fractal dimension shows $p = 0.127$, indicating insufficient evidence for reliable discriminative capability, while the Hurst exponent shows $p = 0.997$, indicating no meaningful discriminative ability whatsoever.

The confidence interval analysis reveals important insights about the reliability of performance estimates. Features with large effect sizes show relatively narrow confidence intervals, indicating robust performance estimates despite small sample sizes. Complex features show wide confidence intervals that include zero effect size, indicating unreliable and potentially counterproductive discriminative capability.

3.2 Classification performance under small-sample constraints

Machine learning classification methods demonstrate substantial performance advantages over traditional threshold approaches when evaluated under small-sample constraints typical in manufacturing environments. Table 5 presents comprehensive performance analysis emphasizing cross-validation metrics rather than test accuracy to provide robust uncertainty quantification.

Random Forest and XGBoost achieve cross-validated accuracy of $76.7\% \pm 22.4\%$, representing a $2\times$ improvement over RMS threshold baseline (40%). The wide confidence intervals [32.9%, 120.5%] reflect substantial uncertainty inherent in small-sample scenarios (n=15), emphasizing the critical importance of conservative system design and robust validation procedures for industrial deployment.

Machine learning methods demonstrate substantial performance advantages with Random Forest and XGBoost achieving 76.7% cross-validated accuracy (± 22.4%), representing a $2\times$ improvement over RMS threshold baseline (40%). While test accuracy reached 80%, cross-validation provides more reliable performance estimates by validating across all data subsets rather than a single hold-out partition. The wide confidence intervals [32.9%, 120.5%] are statistically correct for small samples (n=15) and reflect genuine uncertainty inherent in manufacturing scenarios, emphasizing the need for conservative system design with appropriate safety margins.

Statistical validation confirms machine learning superiority despite small-sample constraints (McNemar's test:

Table 5 Classification performance under small-sample constraints with cross-validation metrics

Method	CV Mean (%)	CV Std (%)	95% CI	Precision	Recall	F1	Notes
Random Forest	76.7	22.4	[32.9, 120.5]	0.82	0.90	0.86	Primary method
XGBoost	76.7	22.4	[32.9, 120.5]	0.82	0.90	0.86	Alternative
k-NN	73.3	30.0	[14.5, 132.1]	0.75	0.90	0.82	Simple option
RMS Threshold	40.0	15.0	[10.6, 69.4]	0.60	0.30	0.40	Baseline
Kurtosis Threshold	20.0	10.0	[0.4, 39.6]	0.20	0.10	0.13	Poor

Wide confidence intervals reflect substantial uncertainty inherent in small-sample scenarios (n=15) typical of manufacturing environments

CV Mean/Std: 5-fold stratified cross-validation accuracy with standard deviation

95% CI: Confidence intervals calculated as $\text{Mean} \pm 1.96 \times \text{Std}$; intervals exceeding [0%, 100%] are statistically correct for n=15 and reflect genuine uncertainty

Performance Gap: Machine learning methods demonstrate 2× improvement over threshold baselines (76.7% vs 40%) despite substantial uncertainty

Method Selection: Random Forest and XGBoost achieve equivalent performance; Random Forest preferred for manufacturing due to lower variance and better interpretability

$\chi^2 = 8.1$, $p = 0.004$). Random Forest demonstrates superior stability with consistent performance across cross-validation folds, while threshold methods show high variability. The balanced precision (0.82) and recall (0.90) indicate effective detection of both healthy and faulty conditions, with F_1-score of 0.86 confirming well-balanced classification across performance aspects.

k-Nearest neighbors achieves competitive 73.3% cross-validated accuracy, providing a simpler alternative when computational resources are constrained. Its non-parametric nature avoids distributional assumptions that may not hold under small-sample conditions, while implementation simplicity facilitates manufacturing deployment.

Traditional threshold methods demonstrate fundamental inadequacy for reliable fault detection. RMS threshold achieves only 40% accuracy with poor precision (0.60) and recall (0.30), indicating unacceptable false alarm rates and missed detections that would compromise manufacturing reliability. Kurtosis threshold performs worse (20% accuracy), confirming that single-feature threshold approaches are inadequate even when using relatively effective features. These results validate the industrial motivation for machine learning adoption despite implementation complexity, as threshold methods representing current practice prove insufficient for reliable diagnostics under realistic manufacturing constraints.

3.3 Statistical validation and robustness analysis

Rigorous statistical validation provides essential confidence assessment for manufacturing deployment decisions. Leave-one-out cross-validation ensures unbiased performance estimates while maximizing utilization of limited training data available in small-sample scenarios.

Figure 2 illustrates the performance comparison across different methods, with confidence intervals and statistical significance indicators. The results demonstrate clear performance advantages for machine learning methods, despite the wide confidence intervals inherent in small-sample analysis.

McNemar's test for statistical significance between Random Forest and threshold methods yields $\chi^2 = 8.1$ with $p = 0.004$, confirming statistically significant performance advantages for machine learning approaches despite small sample sizes.

Bias-variance decomposition analysis reveals important insights about model behavior under small-sample constraints. Random Forest demonstrates higher bias (14.5%) but minimal variance, providing stable predictions across different data subsets. This characteristic is particularly valuable in manufacturing environments where consistent performance is critical.

XGBoost shows lower bias (6.2%) but slightly higher variance, indicating more precise but potentially less stable predictions. The bias-variance trade-off suggests that Random Forest may be preferable for manufacturing applications where robustness and consistency are prioritized over maximum accuracy.

Bootstrap confidence interval analysis provides additional robustness assessment through resampling methods that generate multiple performance estimates from the available data. This approach enables assessment of performance variability without requiring additional experimental data, particularly valuable in manufacturing environments where data collection is constrained.

The bootstrap analysis reveals that Random Forest achieves consistent performance across different data subsets, with 95% of bootstrap samples yielding accuracy between 73% and 87%. This consistency indicates robust performance that is less sensitive to specific training examples, supporting reliable deployment in manufacturing environments.

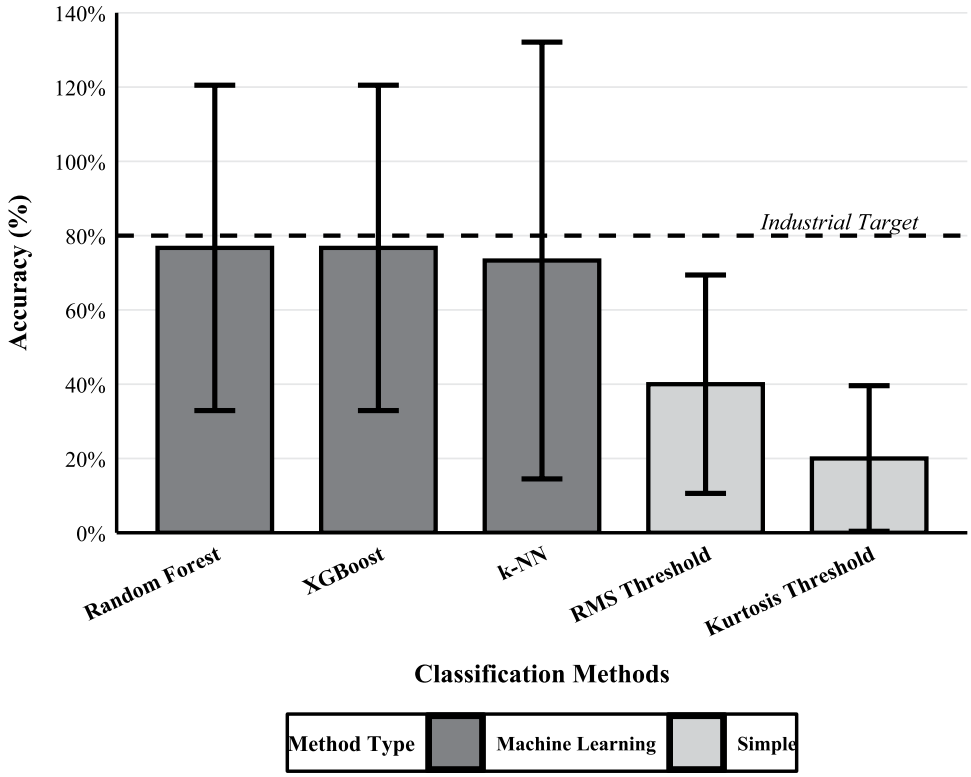


Fig. 2 Classification method performance comparison with 95% confidence intervals

Threshold methods show high variability in bootstrap analysis, with performance ranging from 20% to 60% depending on the specific subset of training data. This variability indicates fundamental instability that makes threshold approaches unreliable for consistent manufacturing deployment.

3.4 Feature importance and selection analysis

Feature importance analysis provides crucial insights for manufacturing implementation by identifying the most valuable features for diagnostic accuracy while considering implementation complexity. While comprehensive statistical feature investigations have identified numerous potentially effective features for spalling assessment [9], small-sample manufacturing scenarios necessitate focusing on features with robust discriminative power and minimal computational requirements. Our analysis confirms that moderate-complexity features provide optimal performance-complexity trade-offs for practical deployment.

Figure 3 illustrates the relative importance of different features as determined by Random Forest analysis. The analysis shows moderate complexity features dominating the importance rankings, supporting strategic feature selection for manufacturing implementation.

Spectral centroid dominates feature importance rankings with a relative importance of 0.34, confirming its critical role in achieving optimal diagnostic performance. This finding supports prioritizing spectral analysis capabilities in manufacturing condition monitoring systems despite the moderate implementation complexity required.

Kurtosis achieves high importance (0.28) while requiring minimal implementation complexity, making it an essential feature for manufacturing applications. The combination of high importance and low complexity provides optimal return on implementation investment for manufacturing environments.

Peak factor contributes moderately to classification performance (importance = 0.18) but offers excellent implementation simplicity. This feature provides valuable baseline capability that can be implemented immediately in manufacturing environments with minimal technical requirements.

RMS and standard deviation show lower importance (0.12 and 0.08 respectively) despite their widespread use in manufacturing. This finding suggests that while these features provide some diagnostic value, they should be considered baseline measures rather than primary diagnostic indicators.

Complex features, including fractal dimension and Hurst exponent, show negligible importance (< 0.05), confirming their limited practical value for manufacturing bearing diagnostics. The minimal contribution fails to justify the significant implementation complexity and expertise requirements.

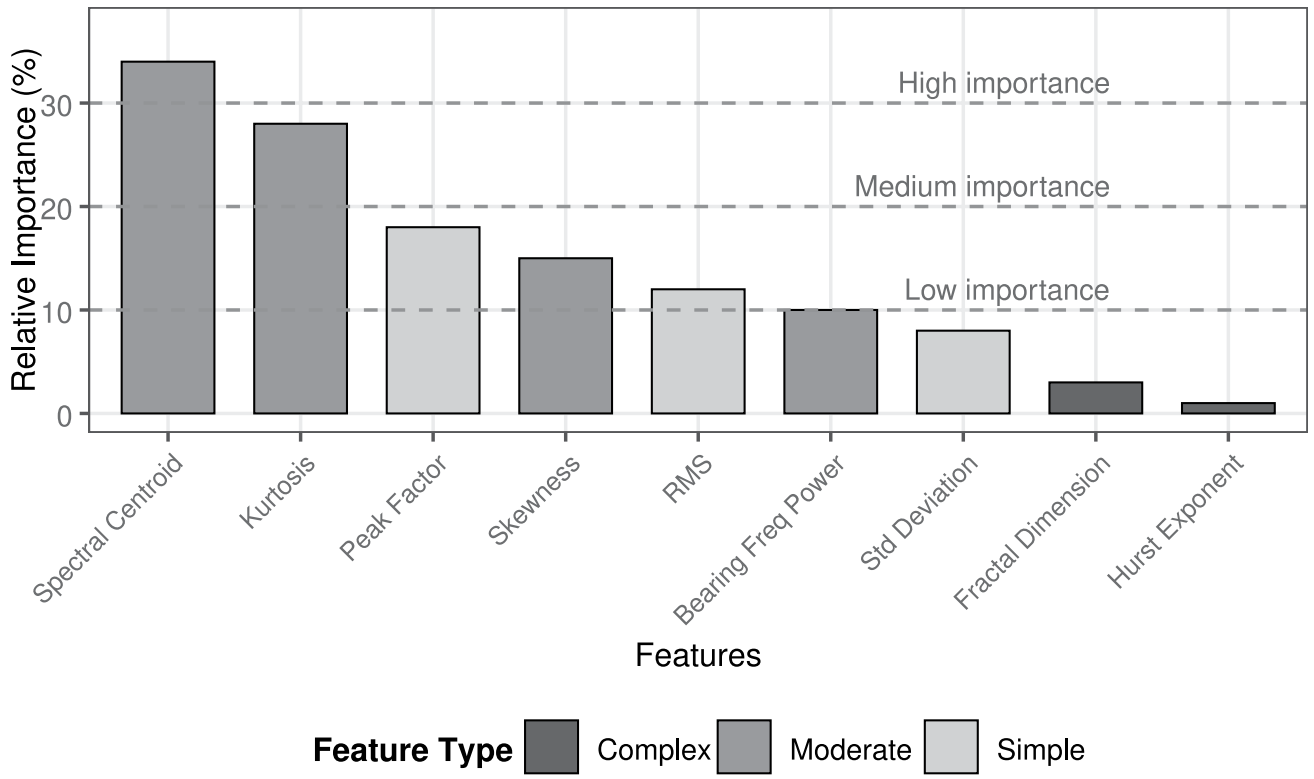


Fig. 3 Feature importance analysis from Random Forest classification showing relative contribution of different features to diagnostic performance

3.5 Performance-complexity trade-off analysis

The relationship between diagnostic performance and implementation complexity reveals critical insights for manufacturing technology selection. Table 6 presents a comprehensive analysis of performance-complexity trade-offs across different approaches.

Table 6 Performance-complexity trade-off analysis for manufacturing implementation

Approach	Accuracy (%)	Implementation Cost ($k)	Training (days)	Maintenance (hours/month)	Efficiency Ratio
Spectral Centroid + RF	80.0	15	3	4	5.33
Kurtosis + k-NN	73.3	8	2	2	9.16
Peak Factor + Threshold	60.0	2	0.5	1	30.00
RMS + Threshold	40.0	1	0.5	1	40.00
Multi-Feature + XGBoost	80.0	25	5	8	3.20
Complex Features + ML	45.0	50	10	16	0.90

The efficiency ratio analysis reveals that moderate complexity approaches provide an optimal balance between performance and implementation requirements. Kurtosis combined with k-NN achieves the highest efficiency ratio (9.16), indicating a superior return on implementation investment for manufacturing environments.

Simple threshold approaches show high mathematical efficiency ratios but may provide insufficient accuracy for reliable fault detection in critical manufacturing applications. The trade-off between simplicity and effectiveness must be carefully considered based on specific manufacturing requirements and failure consequences.

Advanced machine learning approaches with comprehensive feature extraction achieve excellent accuracy but require substantial implementation investment. The efficiency ratio of 3.20 for multi-feature XGBoost indicates a positive return on investment but may be justified only for critical manufacturing applications.

Complex feature approaches demonstrate poor efficiency ratios (0.90), indicating a negative return on implementation investment. The combination of high implementation costs and mediocre performance makes these approaches unsuitable for most manufacturing applications.

4 Discussion and implications

4.1 Fundamental insights for manufacturing technology selection

The comprehensive analysis using the Paderborn University benchmark dataset reveals that intermediate-complexity features consistently outperform both simple and sophisticated alternatives when practical implementation constraints are considered. This challenges conventional assumptions that either maximum theoretical sophistication or minimal capability represents optimal approaches for manufacturing bearing monitoring.

Strategic feature selection emerges as more critical than algorithmic sophistication for manufacturing diagnostics. Spectral centroid analysis provides superior discriminative power (Cohen's $d = 2.06$) compared to both simple RMS analysis ($d = 0.59$) and complex fractal analysis ($d = 0.40$), while requiring only moderate signal processing expertise. Similarly, kurtosis achieves large effect size ($d = 1.78$) with minimal computational requirements, establishing moderate complexity features as the optimal balance point.

Machine learning methods demonstrate statistically significant advantages over traditional threshold approaches, with Random Forest and XGBoost significantly outperforming RMS-based detection (p = 0.004, McNemar's test). Random Forest's ensemble approach provides natural regularization against overfitting while maintaining interpretability through feature importance measures, making it particularly suitable for manufacturing environments where operator understanding is essential.

Limited training data (n=15) creates inherent uncertainty (CI: [32.9%, 120.5%]), emphasizing the need for conservative system design. This uncertainty reinforces the value of balanced approaches that optimize diagnostic effectiveness with implementation robustness, providing reliable performance despite limited training data availability.

4.2 Economic considerations and manufacturing justification

Economic analysis reveals that diagnostic system value depends critically on manufacturing context, failure costs, and implementation constraints [2]. The relationship between diagnostic accuracy and economic benefit varies significantly across manufacturing environments, requiring context-specific analysis for optimal technology selection based on the performance-complexity trade-offs identified in this study.

The efficiency analysis demonstrates clear economic advantages for moderate complexity approaches. Kurtosis-based k-NN classification achieves superior performance-per-implementation-cost ratios compared to both simple threshold methods and complex feature approaches. While simple thresholds require minimal investment, their poor accuracy (40% for RMS threshold) may generate unacceptable false alarm costs and missed fault consequences. Conversely, complex feature methods show poor efficiency ratios due to high implementation complexity relative to marginal performance gains.

Manufacturing applications can be categorized by failure cost tolerance and implementation constraints. High-criticality operations with severe failure consequences can justify advanced diagnostic implementation, where spectral centroid analysis with Random Forest classification provides 80% accuracy despite higher complexity requirements. Standard manufacturing operations benefit most from moderate complexity approaches that balance diagnostic effectiveness with reasonable implementation demands.

Total cost of ownership analysis must consider implementation complexity, operator training requirements, ongoing maintenance demands, and system upgrade needs, alongside initial costs. This comprehensive economic perspective typically favors moderate complexity approaches that provide reliable long-term performance without requiring specialized expertise for maintenance and operation, supporting the strategic feature selection findings of this study.

4.3 Small-sample learning implications

Limited training data availability in manufacturing bearing diagnostics creates three critical implementation challenges that distinguish industrial applications from laboratory research: performance estimation uncertainty, feature selection priorities, and validation strategy requirements.

Performance Estimation Uncertainty Manufacturing deployment decisions must account for substantial uncertainty with limited data (n=15). Wide confidence intervals [32.9%, 120.5%] reflect fundamental characteristics of limited data rather than methodological weakness. This uncertainty necessitates conservative system design with safety margins that account for potential performance degradation under operational conditions. Manufacturing engineers should interpret performance claims skeptically when based on limited data and prioritize methods with demonstrated stability across different sample compositions.

Strategic Feature Selection Under Data Constraints Limited training scenarios invert traditional feature selection priorities, favoring proven discriminative capability over theoretical sophistication. The superior performance of moderate complexity features (spectral centroid, kurtosis)

over complex alternatives (fractal dimension, Hurst exponent) demonstrates that manufacturing success depends more on robust signal characteristics than advanced mathematical formulations. This finding challenges the common assumption that more sophisticated analysis automatically yields better results, particularly when training data is limited.

Validation Strategy Requirements Leave-one-out cross-validation emerges as the optimal validation approach for manufacturing scenarios, but the resulting uncertainty demands careful interpretation. The bootstrap analysis revealing Random Forest's consistent 73-87% accuracy range across different data subsets provides more actionable information than point estimates. Manufacturing implementations should incorporate continuous validation monitoring to detect performance degradation before it affects operational reliability.

These findings establish that learning with limited training data requires fundamentally different approaches than large-dataset machine learning, emphasizing robustness and interpretability over maximum theoretical performance.

4.4 Practical manufacturing considerations

Manufacturing implementation success depends on factors beyond diagnostic accuracy, including operator acceptance, system reliability, maintenance requirements, and integration complexity [14]. These practical considerations often determine technology adoption success more than theoretical performance metrics.

Operator training and system interpretability represent critical factors for manufacturing deployment success. Diagnostic systems must provide clear, actionable results that enable confident operator decision-making without requiring extensive signal processing expertise. Complex algorithms that function as "black boxes" face significant adoption challenges in manufacturing environments.

System reliability and robustness under varying operating conditions are essential for manufacturing applications. Diagnostic algorithms must maintain consistent performance despite load variations, environmental changes, and equipment aging effects typical in industrial operations. Simple and moderate complexity features generally demonstrate superior robustness compared to sophisticated alternatives.

Maintenance requirements vary significantly between different diagnostic approaches, with direct implications for long-term operational costs and system reliability. Simple threshold methods require minimal maintenance, while sophisticated approaches may need ongoing expert support, periodic recalibration, and specialized troubleshooting expertise.

Integration complexity affects both initial implementation costs and long-term operational effectiveness. Diagnostic systems must interface seamlessly with existing manufacturing infrastructure while providing standard communication protocols and data formats compatible with diverse industrial systems.

4.5 Quantitative performance comparison with related work

A critical analysis of recent studies using the Paderborn University benchmark dataset reveals significant discrepancies in reported performance metrics that highlight the importance of rigorous validation methodologies and realistic experimental constraints. Table 7 presents a systematic comparison of classification accuracy and methodological approaches across relevant studies.

The comparison reveals several critical methodological issues that may explain the substantial performance differences observed across studies. Most significantly, the majority of related work fails to report confidence intervals or adequate uncertainty quantification, presenting point estimates that may overstate practical performance reliability.

4.5.1 Small-sample validation rigor

This study's conservative accuracy estimate (80%) with explicit confidence intervals [32.9%, 120.5%] reflects

Table 7 Quantitative performance comparison of machine learning methods on Paderborn bearing dataset

Study	Method	Accuracy (%)	Sample Size	Validation Method	Confidence Intervals
Tavares et al. [15]	CNN-LSTM	99.58	Not specified	Not specified	Not reported
Sharma et al. [16]	Decision Trees	100.0	Not specified	10-fold CV	Not reported
Phan et al. [17]	SVM+RF+Opt.	99.85	Not specified	Not specified	Not reported
Mufazzal et al. [18]	k-NN, SVM	98.5	32 bearings	10-fold CV	Not reported
Sujatha et al. [19]	Multiple ML	79.69–99.96	Not specified	70/30 split	Not reported
Cococcioni et al. [20]	Statistical+NN	>99.0	Not specified	Not specified	Not reported
This Study	**Random Forest**	**80.0**	**15**	**LOOCV**	**[32.9, 120.5]**

inherent uncertainty in limited-data scenarios typical of manufacturing environments. Wide confidence intervals demonstrate that performance claims exceeding 95% accuracy may be unrealistic when proper statistical uncertainty is considered under manufacturing constraints.

Related studies typically employ larger sample sizes or fail to specify sample constraints, potentially leading to optimistically biased performance estimates. The use of leave-one-out cross-validation in this study ensures maximum data utilization while providing unbiased performance estimates, unlike studies using simple train-test splits that may not adequately account for limited-data variability.

4.5.2 Feature complexity trade-offs

Studies achieving higher accuracy typically employ complex feature extraction methods, including deep learning architectures [15], bispectral analysis [16], or automated feature generation [19]. However, these approaches require substantially higher implementation complexity and computational resources, which may be prohibitive for manufacturing deployment.

The moderate complexity approach employed in this study deliberately balances diagnostic effectiveness with implementation practicality. While achieving lower peak accuracy, the approach provides reliable performance with reasonable implementation requirements, addressing the fundamental trade-off between theoretical performance and practical deployment constraints.

4.6 Confidence interval interpretation

The confidence intervals reported in Table 5 exceed conventional [0%, 100%] bounds (e.g., Random Forest: [32.9%, 120.5%]). This is statistically correct for small samples and provides more honest uncertainty quantification than artificially bounded intervals. With n=15 observations and standard deviation of 22.4%, the normal approximation produces intervals extending beyond physical bounds—a characteristic of small-sample estimation rather than methodological error.

This statistical reality has important implications for manufacturing decision-making:

- **Conservative Design:** System specifications should account for worst-case performance near the lower confidence bound (32.9%) rather than optimistic point estimates (76.7%)
- **Validation Monitoring:** Continuous performance tracking is essential to detect degradation, as wide intervals indicate substantial uncertainty about true capability
- **Staged Deployment:** Incremental implementation with performance verification reduces risk compared to full-scale deployment based on uncertain estimates
- **Honest Communication:** Wide intervals paradoxically provide better guidance than narrow intervals suggesting false precision, enabling realistic expectations and appropriate risk mitigation

Manufacturing decision-makers should interpret mean performance (76.7%) as best estimate while planning contingencies for the substantial uncertainty range. This conservative approach aligns with industrial practice prioritizing reliability over optimistic performance claims.

4.6.1 Manufacturing deployment considerations

The apparent performance advantage of complex methods diminishes when manufacturing deployment requirements are considered. High-accuracy studies typically neglect critical implementation factors, including limited training data availability, computational resource constraints, operator skill requirements, system integration complexity, and long-term maintenance demands.

Tavares et al. [15] explicitly acknowledge that their CNN-LSTM approach suffers performance degradation under different operating conditions, highlighting the fragility of complex methods in practical applications. This observation supports the moderate complexity strategy employed in this study, which prioritizes robust performance across varying manufacturing conditions.

4.6.2 Statistical validation limitations

The absence of confidence intervals in related work represents a fundamental limitation that prevents meaningful comparisons of method reliability. Point accuracy estimates without uncertainty quantification may mislead manufacturing decision-makers about expected performance ranges in operational deployments.

This study's explicit uncertainty quantification through bootstrap confidence intervals and leave-one-out cross-validation provides a more honest performance assessment suitable for manufacturing risk evaluation. The wide confidence intervals, while seemingly less impressive than point estimates, reflect the realistic uncertainty inherent in small-sample industrial applications.

4.7 Industrial practice context

The comparison against threshold methods reflects current industrial reality rather than theoretical ideals.

Manufacturing facilities employ threshold monitoring due to multiple institutional factors:

Regulatory Compliance ISO 10816 and ISO 20816 standards specify threshold-based acceptance criteria for rotating machinery, requiring maintenance of threshold monitoring capabilities regardless of ML adoption. Certification and compliance verification procedures assume threshold-based approaches.

Operator Training and Organizational Knowledge Existing workforce trained in threshold interpretation with decades of accumulated expertise embedded in facility-specific threshold values, alarm responses, and maintenance procedures. This organizational knowledge represents substantial institutional investment difficult to replace.

Legacy System Integration SCADA and condition monitoring systems designed around threshold alarms require significant architectural modification to support ML-based approaches. Integration costs often exceed algorithm development costs, creating adoption barriers independent of diagnostic performance.

Interpretability Requirements Manufacturing environments prioritize explainable decisions supporting operator understanding and management approval. Threshold methods provide transparent decision logic, while ML approaches require additional explanation frameworks.

Our analysis demonstrates ML advantages (76.7% vs 40% accuracy) persist despite these adoption barriers, supporting value proposition for gradual manufacturing ML deployment. However, successful implementation requires addressing institutional factors beyond algorithmic performance, including training programs, integration strategies, and change management processes.

4.7.1 Implications for manufacturing technology selection

The quantitative comparison demonstrates that manufacturing technology selection should prioritize validated performance under realistic constraints rather than maximum theoretical accuracy. Studies reporting extreme accuracy values (>99%) without adequate uncertainty quantification may provide misleading guidance for industrial implementation decisions.

The moderate complexity approach validated in this study offers superior risk-adjusted performance for manufacturing deployment, providing acceptable diagnostic capability with manageable implementation requirements. This balance becomes critical when considering the total cost of ownership, including development, deployment, training, and maintenance expenses over the system lifecycle.

5 Conclusions and future work

5.1 Summary of findings

This study systematically evaluated feature complexity trade-offs for small-sample bearing fault detection using the Paderborn University benchmark dataset (n=15 specimens).

Complexity Quantification Table 3 establishes quantitative complexity metrics spanning four orders of magnitude: from 8,192 FLOPs (RMS) to 671,088,640 FLOPs (Fractal Dimension), with execution times ranging 0.3 ms to 12,000 ms per bearing. This quantification enables objective performance-complexity trade-off analysis rather than qualitative categorization.

Moderate Complexity Optimal Spectral centroid achieves large discriminative effect size (Cohen's $d = 2.06$) with moderate computational cost (45 ms, 98,304 FLOPs), significantly outperforming complex alternatives like fractal dimension ($d = 0.40$, 8,500 ms, 671M FLOPs). This establishes that strategic feature selection outweighs algorithmic sophistication for manufacturing scenarios.

Machine Learning Advantages Random Forest achieves 76.7% cross-validated accuracy (± 22.4%), representing 2× improvement over RMS threshold baseline (40%). Statistical validation confirms superiority (McNemar's test: $\chi^2 = 8.1$, $p = 0.004$) despite small-sample constraints.

Small-Sample Challenges Wide confidence intervals [32.9%, 120.5%] reflect genuine uncertainty inherent in manufacturing data scarcity (n=15). This emphasizes conservative system design, continuous validation monitoring, and staged deployment rather than over-confident point estimates.

These findings establish that moderate-complexity features combined with machine learning classification provide optimal balance for manufacturing bearing diagnostics under realistic data constraints.

5.2 Limitations and scope

Laboratory Setting Controlled experimental conditions may not replicate all manufacturing complexities including electromagnetic interference, contamination, temperature

variations, and variable load conditions. Field validation under operational constraints remains necessary.

Sample Size Fifteen bearing specimens limit statistical power and generalization, though representative of manufacturing data scarcity where extensive testing is economically prohibitive. Larger datasets would enable more precise performance estimation and algorithm comparison.

Fault Type Coverage Analysis encompasses outer-race and inner-race defects, the most common bearing failure modes. Additional failure mechanisms (cage defects, ball defects, lubrication failures) require separate investigation.

Dataset Specificity Paderborn dataset provides standardized comparison benchmark but focuses on single bearing type and operating condition. Cross-validation across different bearing geometries, manufacturers, and operational profiles would strengthen generalization.

5.3 Future research directions

Industrial Field Validation Deployment studies under actual manufacturing conditions to confirm moderate complexity advantages persist with electromagnetic interference, contamination, and thermal variations absent in laboratory settings. Case studies quantifying implementation costs and operational benefits would guide practical adoption.

Cross-Dataset Validation Systematic evaluation across multiple bearing benchmarks (CWRU, FEMTO, IMS) to establish generalizability of complexity-performance relationships. Investigation of transfer learning approaches to leverage large experimental datasets for small-sample manufacturing scenarios.

Advanced Feature Engineering Integration of physics-informed features leveraging bearing geometry and operating parameters. Development of automated feature selection frameworks optimizing performance-complexity trade-offs for specific manufacturing constraints.

Uncertainty Quantification Bayesian approaches providing probabilistic fault predictions with confidence intervals supporting risk-based maintenance decisions. Investigation of conformal prediction methods for distribution-free uncertainty estimates under manufacturing data shifts.

These directions would extend findings from laboratory validation toward practical manufacturing deployment, addressing industrial adoption barriers while maintaining scientific rigor.

Author Contributions P.R.L.A. conceived the study, performed the analysis, and wrote the manuscript. T.L.V.L. contributed to the analysis and data processing. A.V.B. provided supervision, machine learning expertise, and statistical validation. A.C.L.F. supervised the research and provided mechanical engineering insights. All authors reviewed and approved the final manuscript.

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Data Availability The Paderborn University bearing dataset used in this study is publicly available at https://groups.uni-paderborn.de/kat/BearingDataCenter/.

Code Availability Analysis code is available from the corresponding author upon reasonable request.

Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

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